Project

Control an LED connected to LPC1768 (P1.28) based on a field1 value (0 or 1) received from Thingspeak channel using ESP01 wifi module

Block diagram:-

LPC1768 (UART1 TX/RX)

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ESP-01 (AT commands over UART)

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(wifi)

Thingspeak (cloud)

↕ (mobile/web)

user sets field1 to 1 or 0

Uart1

P0.15 - TX

P0.16 - RX

P0.2 - TX

P0.3 RX

Step by Step Algorithm -

1. Initialize UARTs and GPIO
   - uart0 is for debug prints via USB-serial to your PC
   - uart1 is for communicating with ESP01
   - set GPIO pin1.28 as output for the LED
2. ESP01 initialization via AT commands

Command	Purpose
AT	check if ESP01 is responding
AT+CWMODE=1	set ESP to station (client mode)
AT+CWJAP="SSID","password"	connect to your wifi

Responses like "OK" or "wifi connected" are parsed in wait_response() function

3. Fetch Data from thingspeak

Every 2 seconds inside main() loop:-

a) open TCP Connection

AT + CIPSTART = "TCP", "api.thingspeak.com", 80

b) send HTTP GET REquest

c) send data length

4) Parse the Json Response